Climate change impact on future **wildfire** danger and activity in southern Europe: a **review**

J Dupuy, H Fargeon, N Martin-StPaul, F Pimont… - Annals of Forest …, 2020 - Springer

… moisture, which in turn conditions fire **spread** in the fire module. Both LPJ-… **spread** in SPITFIRE, while the maximum daily Nesterov index is used to calibrate the annual burnt area **model** …

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Rapid **prediction** of **wildfire spread** using ensemble Kalman filter and polyline simplification

S Yoo, J Song - Environmental Modelling & Software, 2023 - Elsevier

… This paper proposes a new method for rapid **prediction** of **wildfire spread**, which employs computational **wildfire** simulations by FARSITE and assimilates the **simulation** results with …

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Wildfire danger **prediction** and understanding with deep learning

S Kondylatos, I Prapas, M Ronco… - Geophysical …, 2022 - Wiley Online Library

… of pivotal importance in modeling **wildfire dynamics**. To test the … that influence the occurrence and **spread** of wildfires and … of the danger **maps** produced by the LSTM **model**. Figure 2a …

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Wildland fire prediction based on statistical analysis of multiple solutions

G Bianchini - 2006 - ddd.uab.cat

… Once all this process is finished, we are in front of a **forest fire spread** simulator. In this … for each point in the **map** and each time interval in the **simulation**. This resulted in millions of …

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Predictive modelling of fire occurrences from different fire **spread** patterns in Mediterranean landscapes

A Duane, M Pique, M Castellnou… - … Journal of **Wildland Fire**, 2015 - CSIRO Publishing

… allowed the **prediction** of occurrences from different fire **spread** patterns in … **maps** for the **model** for each decade and its projection to the other decade for each kind of fire **spread** …

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Genetic and firefly metaheuristic algorithms for an optimized neuro-fuzzy **prediction** modeling of **wildfire** probability

A Jaafari, SVR Termeh, DT Bui - Journal of environmental **management**, 2019 - Elsevier

… The efficiency of these models was compared to a single ANFIS **model** and … **wildfire prediction** accuracy compared to the single ANFIS **model** (AUC success rate = 0.82; AUC **prediction** …

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A **survey** of machine learning algorithms based forest fires **prediction** and detection systems

F Abid - Fire technology, 2021 - Springer

… The **forest fire behavior prediction** concerns the level of … to construct the **Forest-Fire Susceptibility** (FFS) **map** of the … In this study a **model** for **wildfire** ignition probabilities **prediction** …

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Developing the US **wildland fire** decision support system

EK Noonan-Wright, TS Opperman… - Journal of …, 2011 - Wiley Online Library

… , fire **behavior prediction**, economic assessment, smoke … The objective of this paper is to **review** the US **Wildland Fire** … paper **maps** and lengthy land, resource, and fire **management** …

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Studying **wildfire behavior** using FIRETEC

R Linn, J Reisner, JJ Colman… - … journal of **wildland fire**, 2002 - connectsci.au

… **model** capable of describing **wildfire behavior** in a wide variety of circumstances. We believe that a three-dimensional **wildfire model** … done extensive fuels inventory **mapping** as a part of …

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Getting ahead of the **wildfire** problem: Quantifying and **mapping management** challenges and opportunities

CD O'Connor, MP Thompson, F Rodríguez y Silva - Geosciences, 2016 - mdpi.com

… In this **review**, we provide an overview of some of the … Fire **mapping** is implemented in the minimum travel time-based deterministic fire **behavior model** FlamMap5 [35], producing a **map** …

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Guidance on spatial **wildland fire** analysis: models, tools, and techniques

RD Stratton - 2006 - books.google.com

… fire **behavior** advances, so too will our fire **management** … FlamMap (version 3.0) is a spatial fire **behavior mapping** and … Unlike FARSITE, Flam **Map** makes independent fire **behavior** …

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Defining extreme **wildfire** events: Difficulties, challenges, and impacts

F Tedim, V Leone, M Amraoui, C Bouillon… - Fire, 2018 - mdpi.com

… on measurable fire **spread** and **behavior** parameters and … This paper will first present a **review** of the terms used to label … organizations concerned with **wildfire management** but also with …

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Data-driven **wildfire spread** modeling of european wildfires using a spatiotemporal graph neural network

M Rösch, M Nolde, T Ullmann, T Riedlinger - Fire, 2024 - mdpi.com

… DL-based **wildfire spread model** to predict the daily **spread** of … uses a STGNN to **model** the **wildfire spread** for the first time. … Therefore, satellite-derived burned area **mapping** products …

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Wildfire mitigation plans in power systems: A literature **review**

DAZ Vazquez, F Qiu, N Fan… - IEEE Transactions on …, 2022 - ieeexplore.ieee.org

… **wildfire spread simulation** in their risk prioritization decision making [93]. The **wildfire spread simulation** … , a time visualization of the **wildfire** research bibliographic **mapping** is developed. …

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Setting **wildfire** evacuation triggers by coupling fire and traffic **simulation** models: a spatiotemporal GIS approach

D Li, TJ Cova, PE Dennison - Fire technology, 2019 - Springer

… We computed the four ETEs for each **simulation** and **mapped** out the trigger buffers for the … better reveal the **dynamics** of evacuation traffic and fire **spread** in **wildfire** evacuations when …

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Distributed **wildfire** surveillance with autonomous aircraft using deep reinforcement learning

KD Julian, MJ Kochenderfer - … of Guidance, Control, and **Dynamics**, 2019 - arc.aiaa.org

… In addition, an accurate **wildfire simulation model** is required to develop and test … In addition, the **trajectory** is planned using the true **wildfire map** rather than an observation or belief **map**, …

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Wind–terrain effects on the **propagation** of wildfires in rugged terrain: fire channelling

JJ Sharples, RHD McRae… - … Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… Implications of fire channelling for **bushfire** risk **management** are also … (a) **Map** of south-eastern Australia showing the general location of the events considered in the study; (b) **map** …

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A new algorithm for simulating **wildfire spread** through cellular automata

GA Trunfio, D D'Ambrosio, R Rongo… - … Computer **Simulation** …, 2011 - dl.acm.org

… for **wildfire simulation** through Cellular Automata (CA), which is able to effectively mitigate the problem of distorted fire shapes. Such a result is obtained allowing **spread** directions that …

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A **review** of recent forest and **wildland fire management** decision support systems research

DL Martell - Current Forestry Reports, 2015 - Springer

… Fire behaviour or the physics and chemistry of fire ignition and **spread**. … the use of drones to **map** large fires from the restricted air space fire **management** agencies impose over large …

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A **review** of approaches to estimate **wildfire** plume injection height within large-scale atmospheric chemical transport models

R Paugam, M Wooster, S Freitas… - … Chemistry and Physics, 2016 - acp.copernicus.org

… we cannot resolve the plume **dynamics**, parameterizations are … The aim of this manuscript is to **review** the different … of the Sofiev **model** against (i) the original version of the **model**, (ii) …

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